

4(a)	<i>any three from:</i> right ventricle pumps blood to lungs (and back to heart) <u>and</u> left ventricle pumps blood to (rest of) body (and back to heart) / AW ; (so) short(er) / less, distance (for blood to travel) / ora ; less <u>resistance</u> (to, flow / overcome) / ora ; less force / lower pressure, required / ora ;	3
4(b)	<i>any two from:</i> ventilation / breathing (movements) / inspiration / inhalation, brings in oxygen ; blood flow (in pulmonary capillaries) removes oxygen ; oxygen binding to haemoglobin / oxyhaemoglobin formation, (removes oxygen) ; deoxygenated blood arriving / blood arriving low in oxygen ;	2

Question	Answer	Marks
4(c)(i)	any one from: <i>in context of, airways / trachea / bronchi:</i> <i>less air because:</i> diameter of lumen decreases / ref. to smooth muscle contraction ; inflammation ; thicker layer of mucus ; smoke present in inhaled air (so lower proportion of air) ; AVP ;	1
4(c)(ii)	any two from: carbon monoxide (present in smoke) binds to haemoglobin ; ref. to competitive / permanent / irreversible, binding ; presence of carbon monoxide (from smoke) lowers affinity of haemoglobin for oxygen ; ref. to carboxyhaemoglobin formed ; comparatively less haemoglobin per red blood cell to bind oxygen ;	2

Question	Answer	Marks
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